higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

9 December 2025

This marking guideline consists of 6 pages.

tvetpro.co.za



Try our Apps on Play Store

NATED Question Papers	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	**N4** Mechanical Draughting NATED
N1-N3 Electrical Trade Theory NATED	**N2-N3** Diesel Trade Theory NATED	**N5-N6** Strength of Materials & Structures NATED	**N3** Building & Civil Technology NATED	**N1-N2** Fitting and Machining Theory NATED	**N1-N4** Plating and Structural Steel Drawing NATED
N1-N3 Motor Trade Theory NATED	**N5-N6** Mechanical Drawing & Design NATED	**N1-N2** Plumbing Theory NATED	**N4-N6** Fault Finding & Protective Devices NATED	**N2-N3** Instrument Trade Theory NATED	**N5-N6** Engineering Physics NATED
N5-N6 Fluid Mechanics NATED	**N4-N6** Supervisory Management NATED	**N4-N6** Digital Electronics NATED	**N6** Control Systems NATED	**N1-N2** Bricklaying &Plastering Theory NATED	More…

QUESTION 1

1.1
1.1.1 C✓
1.1.2 B✓
1.1.3 B✓
1.1.4 B✓
1.1.5 D✓

(5 × 1) (5)

1.2
1.2.1 C✓
1.2.2 A✓
1.2.3 E✓
1.2.4 B✓
1.2.5 D✓

(5 × 1) (5)
[10]

QUESTION 2

2.1 4✓ (1)

2.2 Self-complement is 1's complent of any number of the code representing the 9's complement of the decimal number.✓✓ (2)

2.3 68, 7_{10} to excess -3

6	8,	7
+3	+3	+3✓
9	11	10
1001	1011,	1010

10011011,1010xs-3✓✓ (3)

2.4 2.4.1 $15,6_8$ = $1101,110_2$ ✓ & 12, 125_{10} = $1100,001_2$ ✓

2's compliment of 12, 125_{10} = 1100,001 + 0011, 110+1 = 0011,111✓

	0	*	1	1	0	1,	1	1	0
	1	*	0	0	1	1,	1	1	1
1	0	*	0	0	0	1,	1	0	1 ✓

The carry is ignored ✓

$1,101_2$ = $1,625_{10}$ ✓ (6)

tvetpro.co.za

2.4.2 $92,5_{10} = 1011100,1_2$ ✓ & $3,C_{16} = 0011,1100 = 11,11_2$ ✓

						1	**1**	**0**	**0**	**0,**	**1**	**0**	**1**	✓
1	1	1	1	1	0	1	1	1	0	0	1	0,	0	0	0	
					1	1	1	1	↓	↓	↓	↓	↓	↓	↓	
				1	0	0	0	0	↓	↓	↓	↓	↓	↓	↓	
					1	1	1	1	↓	↓	↓	↓	↓	↓	↓	✓
					0	0	0	1	0	1	0	0	↓	↓		
									1	1	1	1	↓	↓		
									0	1	0	1	0	0		
											1	1	1	1		✓
													1	0	1	

$11000,101_2 = 30.5_8$ ✓ (6)

2.4.3 $6, D_{16} = 1100, 1101_2$ ✓ & $7, 6_8 = 111, 11_2$ ✓

					1	1	0	0	1	1	0	1		
								1	1	1	1	1		✓
					1	1	0	0	1	1	0	1		
				1	1	0	0	1	1	0	1			✓
			1	1	0	0	1	1	0	1				
		1	1	0	0	1	1	0	1					
	1	1	0	0	1	1	0	1						✓
1	1	0	0	0	1	1	0	1	0	0	1	1		

$1100011,010011_2 = 63, 4C_{16}$ ✓ (6)

[24]

QUESTION 3

3.1 3.1.1 XOR or Exclusive OR ✓ (1)

3.1.2 $\overline{\overline{B}C}$ (1)

3.1.3 $\overline{A \oplus B + \overline{\overline{B}C} + D}$ (3)

3.2 3.2.1 $B\overline{C}\,\overline{D} + \overline{B}\,\overline{C}\,D + BCD + \overline{B}\,C\,\overline{D} = F$

$B(\overline{C}\,\overline{D} + CD) + \overline{B}(\overline{C}\,D + C\,\overline{D})$ ✓
$B(\overline{C \oplus D}) + \overline{B}(C \oplus D)$
$LET\ \ C \oplus D = K$ ✓
$B\overline{K} + \overline{B}K$ ✓
$B \oplus K$ ✓
$B \oplus C \oplus D$ ✓ (6)

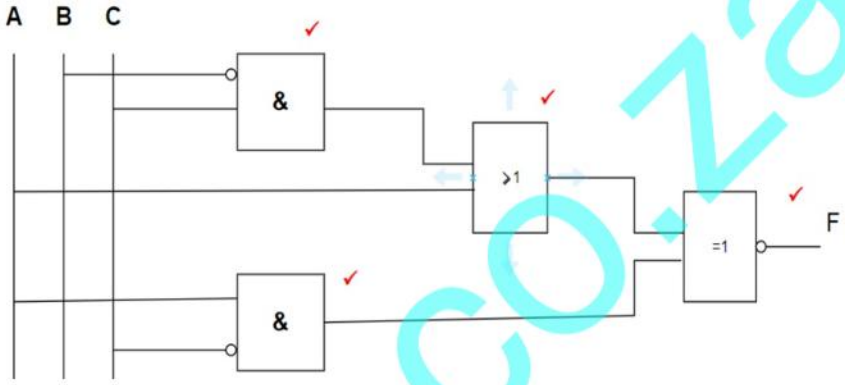
tvetpro.co.za

3.2.2 $AB+\overline{\overline{A}+B}+\overline{A\,\overline{B}\,C+\overline{A}\,BC}=F$

$AB+\acute{A}\ .\overline{B}+(\overline{A}+\acute{B}+\overline{C})((\acute{A}+\overline{B}+\overline{C}$ ✓
$AB+A\,\overline{B}+\overline{A}\,\overline{B}+\overline{A}\,\overline{C}+AB+B\overline{C}+A\,\overline{C}+\overline{B}\,\overline{C}+\overline{C}$ ✓
$A+\overline{A}\,\overline{B}+\overline{C}(\overline{A}+B+A+\overline{B}+1)$ ✓
$A+\overline{A}\,\overline{B}+\overline{C}$ ✓
$A+\overline{B}+\overline{C}$

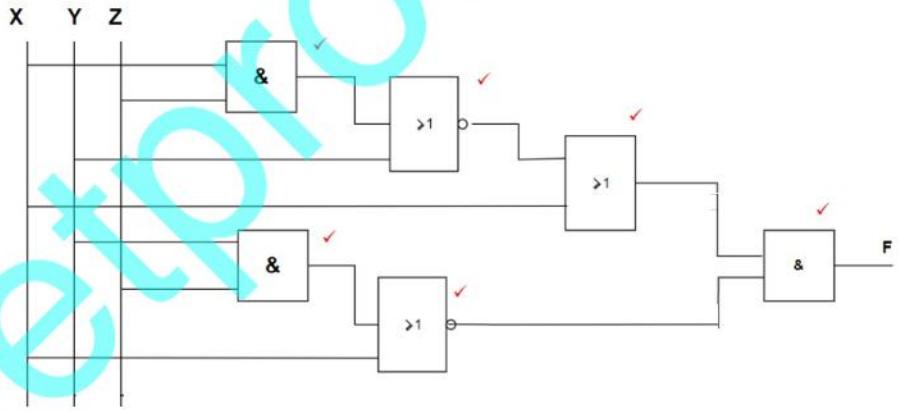
(5)

3.3 3.3.1 $\overline{A+\overline{\overline{B}}C\oplus A\overline{\overline{C}}}=F$



(4)

3.3.2 $(X+\overline{Y+XZ})(\overline{YZ+X})=F$

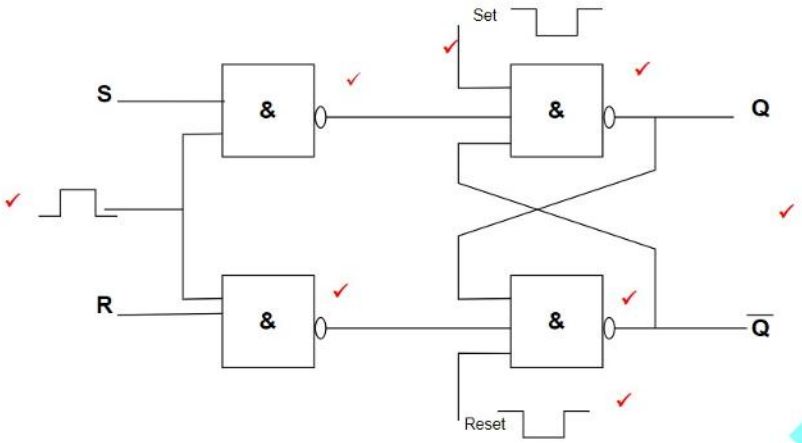


(6)
[26]

tvetpro.co.za

QUESTION 4

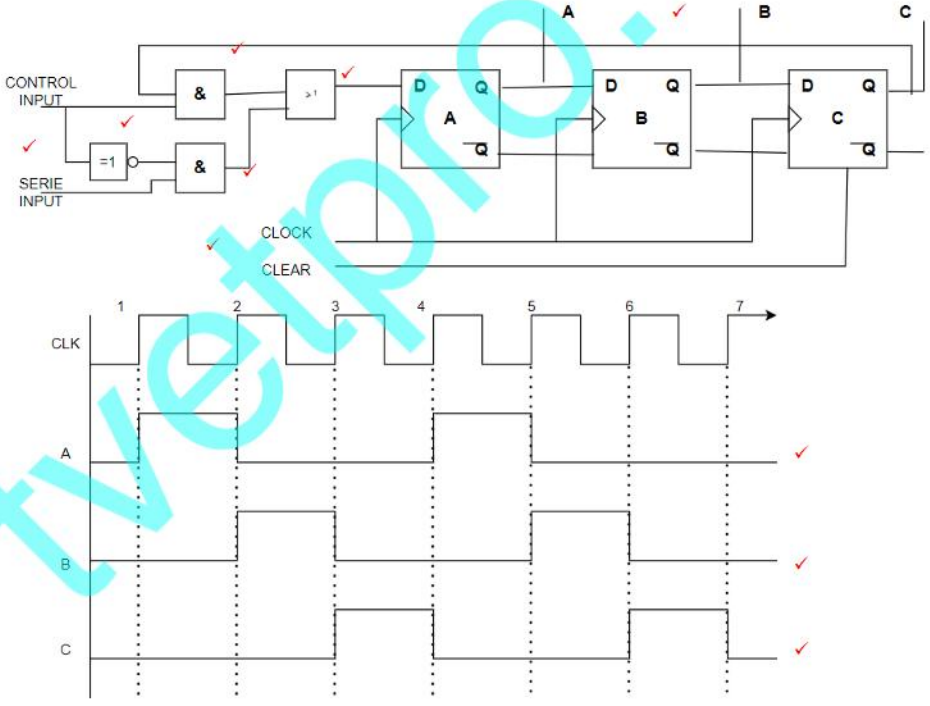
4.1



S	R	Q	$\overline{Q}$	
0	0	Qn	$\overline{Qn}$	NO CHANGE
0	1	0	1	
1	0	1	0	
1	1	1	*	INVALID

(10)

4.2

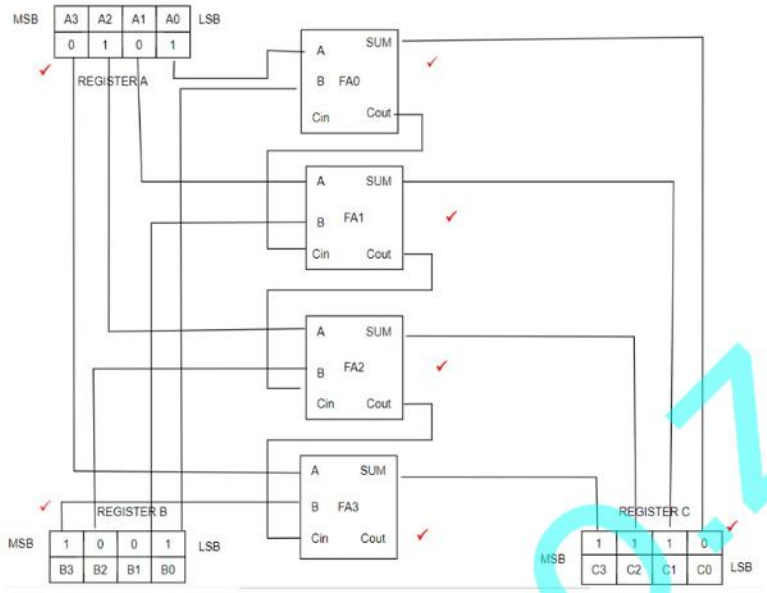


(10)
[20]

tvetpro.co.za

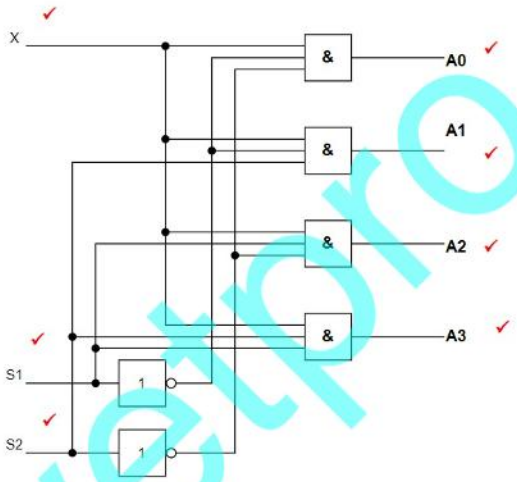
QUESTION 5

5.1



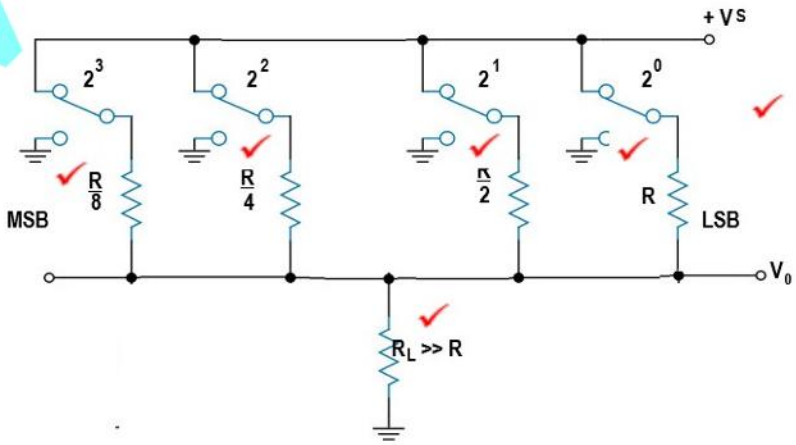
(7)

5.2



(7)

5.3



(6)

[20]

TOTAL: 100

tvetpro.co.za

Copyright reserved